Speech Biomarkers for Parkinson's Disease Screening

Technology assessment and venture whitepaper (research stage)

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Evidence base: frozen protocol FINAL_V1 (audited) · 2026-10-20

Contents

1 Executive summary 2

2 Problem and market 2

3 Product concept 3

4 Technology 4

5 Data asset and its limits 5

6 Evaluation design and integrity 5

7 Results: metrics by feature combination and model 6

8 Every acoustic feature 10

9 Feature families, statistics and feature selection 12

10 Feature importance 14

11 Hyperparameter tuning 15

12 Which speech tasks carry the signal 17

13 The confounding problem - evidence and implications 18

14 Error analysis 18

15 De-risking programme and validation plan 19

16 Regulatory pathway (to be confirmed with regulatory counsel) 20

17 Intellectual property and freedom to operate 20

18 Competitive landscape 20

19 Business model hypotheses (to be tested) 21

20 Risk register 21

21 Team and use of funds 21

22 Conclusion 21

A Audit summary 21

B Glossary 22

Status and reading guide

Research-stage technology assessment prepared for investor due diligence. The product described is **not a medical device** and has **not** been clinically validated. All performance numbers come from one frozen, audited internal study (protocol FINAL_V1, 47 speakers, one Italian corpus). Market and regulatory statements are cited to public sources (reference list at the end) and must be re-verified before any filing or investment decision.

1. Executive summary

The opportunity. About 11.8 million people lived with Parkinson's disease (PD) worldwide in 2021 [1], nearly one million in the United States with about 90,000 new US diagnoses per year [2], and the US economic burden was estimated at about $52 billion per year [3]. Clinical diagnosis is wrong in a meaningful minority of cases, especially early and outside specialist centres (pooled clinicopathologic accuracy about 74-80%) [4]. Speech is affected in about 89% of people with PD [5], it is cheap to record on any phone, and it can be repeated at home. A validated speech screen could triage referrals, support earlier diagnosis and provide digital endpoints for clinical trials. **What was built.** A speaker-independent pipeline that turns a short multi-task speech recording (about 5.4 minutes of median task durations) into a PD probability: 40 hand-crafted acoustic features and three frozen self-supervised speech encoders (WavLM-large, wav2vec2-large-lv60, XLS-R 300M; about 315 million parameters each), four classifier families, feature selection and eight hybrid representations - 56 configurations, each evaluated with nested, speaker-disjoint five-fold cross-validation under a frozen, hash-locked protocol and an independent audit. **What the data show.** On 47 speakers (25 PD, 22 age-comparable controls; 785 recordings) the median pooled speaker-level AUROC over the 56 configurations is 0.995 (range 0.867 to 1.000); 40 of 56 make at most one speaker error. The reference model (WavLM-large + logistic regression, 1025 trainable parameters) reaches AUROC 0.995 [95% CI 0.977-1.000], sensitivity 0.96 (exact 95% CI 0.80-1.00) and specificity 1.00 (exact CI 0.85-1.00).

The critical caveat (read before the numbers)

In this corpus every recording session contains only one class (patients and controls were recorded on different days; part of the patient audio was recorded at a different sampling rate). Four simple statistics of the zero-crossing rate already give AUROC 0.989, the most important features are statistics of the **quietest 10% of frames** (background), and the same representations can tell recording sessions apart **inside one class** for unseen speakers (controls early versus late March: AUROC up to 0.93). The near-perfect separation is therefore **not evidence of disease detection**; it is at least partly a recording-condition effect. This is the single largest technical risk and the first thing a funded programme must remove.

Investment thesis. The asset today is a rigorously engineered and audited evaluation pipeline, a feature-level understanding of where the signal comes from, and a concrete plan to obtain confound-free evidence. It is **pre-validation (proof of concept on a confounded dataset)**. A staged programme (section 15) converts it into an investable clinical asset only if a recording-matched, multi-site cohort reproduces the separation with an AUROC lower confidence bound above 0.80 while a background-only model stays at chance. Milestones are designed so that each tranche buys a decisive answer.

What we can claim	What we cannot claim
A leakage-controlled, reproducible pipeline (280 runs, 20 audit checks, independent re-implementation).	That the models detect Parkinson's disease.
Near-perfect speaker-independent separation on this corpus with simple and with large models.	Any clinical accuracy, sensitivity or specificity for a real screening population.
Which features, tasks and models carry the separation, and how fragile tuning is.	Generalisation to other languages, devices, rooms or disease stages.
Low compute cost per test (CPU seconds).	Regulatory readiness, reimbursement or clinical utility.

2. Problem and market

- Prevalence: about 11.77 million people with PD worldwide in 2021, age-standardised prevalence 138.6 per 100,000, rising (GBD 2021 based analysis) [1].
- United States: nearly 1 million people living with PD (earlier estimates about 650,000) and about 90,000 new diagnoses per year [2]; total economic burden about $52 billion per year (about $25.4 billion direct medical, $26.5 billion non-medical) [3].

- Diagnostic gap: pooled clinicopathologic accuracy of the clinical diagnosis about 74-80%, lower in early disease and with non-specialists; in a 10-year cohort 10.6% of diagnoses were revised [4].
- Speech: about 89% of people with PD have a speech or voice disorder, but only 3-4% receive speech treatment [5]; voice is therefore both a candidate biomarker and an under-served symptom.
- Where a speech screen could create value: (1) triage of referrals to movement-disorder clinics; (2) case finding in primary care or tele-medicine; (3) remote monitoring and digital endpoints in clinical trials; (4) speech-therapy follow-up.

Why now. (1) Self-supervised speech foundation models (wav2vec 2.0, WavLM, XLS-R) give strong frozen representations without disease labels [11]; (2) smartphones make repeated, remote recordings cheap; (3) regulators have begun to process speech-based software devices (section 16); (4) public corpora such as NeuroVoz [8] allow frozen external tests.

Table 1: Illustrative, assumption-driven sizing of one segment only (remote monitoring of people already diagnosed in the US; base of about 1 million people [2]). Adoption and annual price per patient are assumptions to be tested, not forecasts.

Assumed adoption	Assumed annual price	Patients	Annual revenue
1%	$100	10,000	$1.0 M
1%	$300	10,000	$3.0 M
5%	$100	50,000	$5.0 M
5%	$300	50,000	$15.0 M
10%	$100	100,000	$10.0 M
10%	$300	100,000	$30.0 M

Why predictive value matters commercially. A screen is used where prevalence is low. With sensitivity 0.96, a positive result means PD in only about 5% of cases at 1% prevalence if specificity is 0.80, 16% at specificity 0.95 and 49% at specificity 0.99 (figure below). Any go-to-market therefore needs either very high specificity or an enriched population (for example people already referred for motor symptoms).

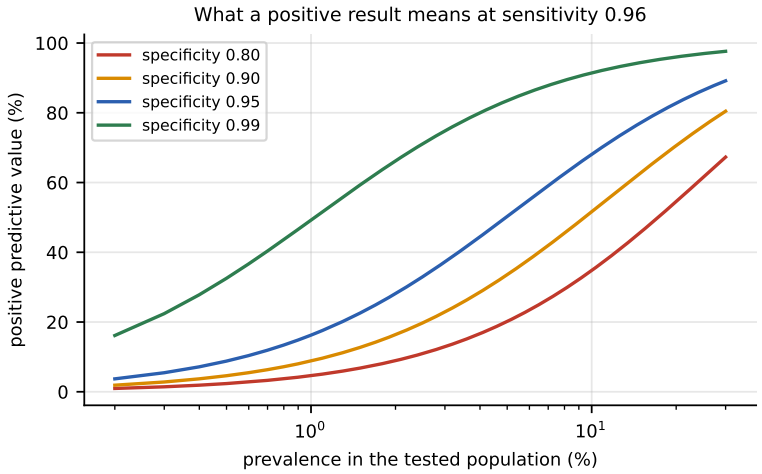


Figure 1: Positive predictive value as a function of prevalence for sensitivity 0.96 and four specificities (formula, no study data).

3. Product concept

A guided recording session on a smartphone or web browser (sustained vowels, /pa/ and /ta/ repetitions, a reading passage, a word list and a phrase; median durations in this corpus give about 5.4 minutes of speech), automatic recording-quality checks, server-side or on-device scoring and a clinician report with a risk score and confidence. The recording-quality and **background-consistency gate** is part of the core product, not an add-on: section 13 shows that background differences can dominate the score.

Table 2: Recording protocol in the corpus (16 task codes; median seconds per recording).

Task code	Recordings	Median s
B1	49	69.7
B2	44	61.9
D1	50	6.5
D2	50	6.2
FB1	47	30.3
PR1	50	55.8
VA1	50	12.3
VA2	49	6.0
VE1	50	12.4
VE2	49	6.1
VI1	50	13.7
VI2	49	6.2
VO1	50	13.2
VO2	49	6.2
VU1	50	12.9
VU2	49	6.4

4. Technology

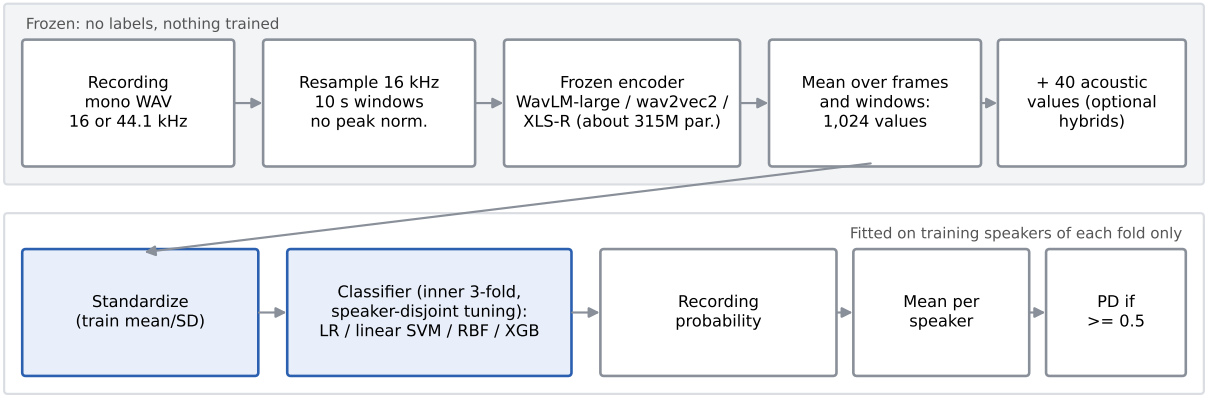


Figure 2: Pipeline: frozen feature extraction (no labels) and the only fitted stages (standardisation and classifier), fitted on training speakers only.

Representations. (i) 40 acoustic features: 10 frame features (log energy, zero-crossing rate, spectral centroid, roll-off at 85%%, spectral flatness, log energy in 0-0.5, 0.5-1, 1-2, 2-4 and 4-8 kHz) x 4 statistics over 25-ms frames (mean, SD, 10th and 90th percentile); (ii) frozen encoders: each 10-s window gives a 499 x 1,024 output map, averaged over time and windows to 1,024 values; (iii) hybrids: concatenations up to 3,112 values. **Models.** Logistic regression (20 configurations), linear SVM (10), RBF SVM (24), XGBoost (24 of a 256-point grid), class weighting inside every grid. **Decision.** Speaker score = mean recording probability, fixed threshold 0.5.

Table 3: Frozen encoders: size, CPU cost per 10-s window (1 / 4 / 8 threads, desktop CPU) and licence (model hub metadata).

Key	Checkpoint	Parameters	Weights MB	s per window	Licence
WLM	microsoft/wavlm-large	315453120	1262	4.96 / 1.63 / 1.52	not declared on model card
W2V	facebook/wav2vec2-large-lv60	315438720	1270	4.39 / 2.21 / 1.42	Apache-2.0
XLSR	facebook/wav2vec2-xls-r-300m	315438720	1270	4.67 / 2.39 / 1.40	Apache-2.0

Cost per test. The corpus averages 45.9 encoder windows per speaker; at 1.52 s per window with 8 threads the WavLM pass costs about 70 s of wall time (about 0.15 core-hours) per test. The 40 acoustic features take 0.08 s per minute of audio on one thread. At an assumed cloud price of p dollars per vCPU-hour the encoder cost per test is about 0.15 x p dollars (for example about $0.008 at an assumed p = $0.05), i.e. compute is not a cost driver; clinical validation, regulatory work and data acquisition are.

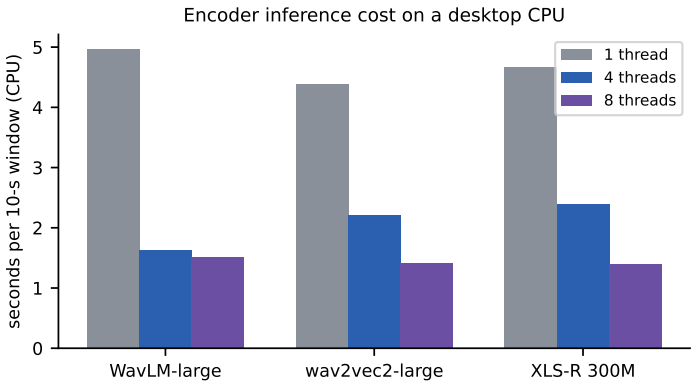


Figure 3: Measured CPU inference cost of the three encoders.

5. Data asset and its limits

Italian Parkinson's Voice and Speech [6] (IEEE DataPort, listed as Creative Commons Attribution [7]): after exclusions **785 recordings, 47 speakers (25 PD, 22 elderly controls), 4.98 audio hours, 16 task codes**. Spreadsheet demographics: PD 66.8 +/- 9.1 years (40-80), 16 men / 9 women; controls 67.1 +/- 5.2 years (60-77), 10 men / 12 women. Exclusions: 15 young controls (age confound) and one byte-identical duplicate recording; three people recorded in two sessions were merged into one speaker each. Four controls are flagged in the corpus spreadsheet for poor recording quality or reading difficulty.

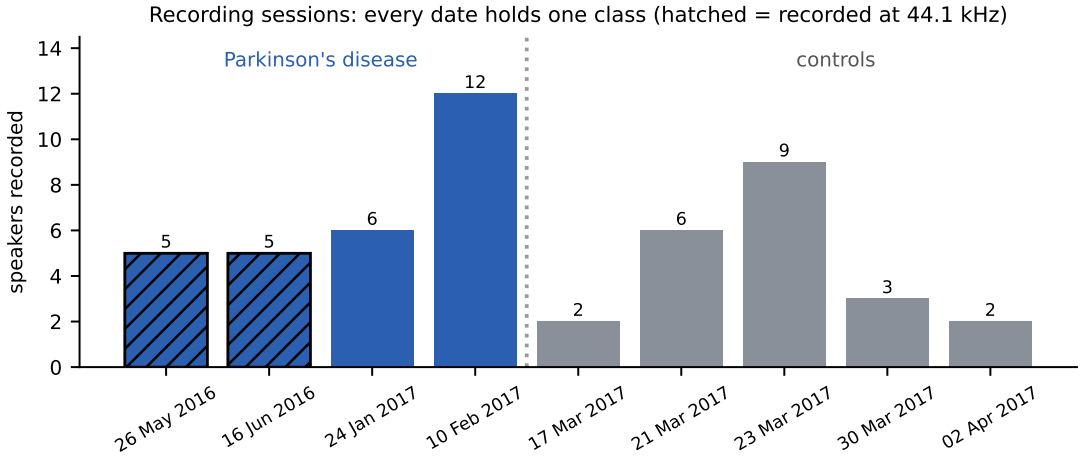


Figure 4: The structural confound: recording dates, class and sampling rate. No date contains both classes; all 44.1-kHz audio is PD.

- No disease severity, duration or medication data; no clinical scale (for example MDS-UPDRS) to relate scores to disease stage.
- Single site, single language, small sample: one speaker changes a speaker-level metric by about two percentage points.
- Sex balance differs between groups (PD 64% men, controls 45% men); age distributions overlap (60-80 for most).
- The dataset licence permits attribution-based reuse, but clinical-grade development requires new, consented, protocol-controlled data (section 15).

6. Evaluation design and integrity

- Frozen protocol (SHA-256 5535d1a440a57005...) written before any model saw the data; 56 conditions x 5 outer folds = 280 runs; FWL adjustment blocked because it was not defined in advance.

- Stratified, speaker-disjoint 5-fold outer CV; nested speaker-disjoint 3-fold inner CV for tuning and feature selection; every learned step fitted on training speakers only; fixed threshold 0.5.
- Integrity gates on every run (speaker overlap, coverage, probability validity, labels, folds, hashes) and an out-of-fold audit: 20 of 20 checks passed, 0 failed runs.
- Independent audit (this document's appendix): independent re-parse of the raw data, re-extraction of embeddings (max difference 3.1e-06), independent nested CV (same selected configurations), recomputation of all metrics (0 disagreements), bit-identical re-execution of all 70 logistic-regression runs under single-threaded BLAS. No result-changing bug; documentation errata recorded.
- Primary metrics: pooled speaker-level AUPRC and AUROC (prevalence 53.2% PD); supporting: F1, sensitivity, specificity, balanced accuracy, bootstrap and exact intervals.

7. Results: metrics by feature combination and model

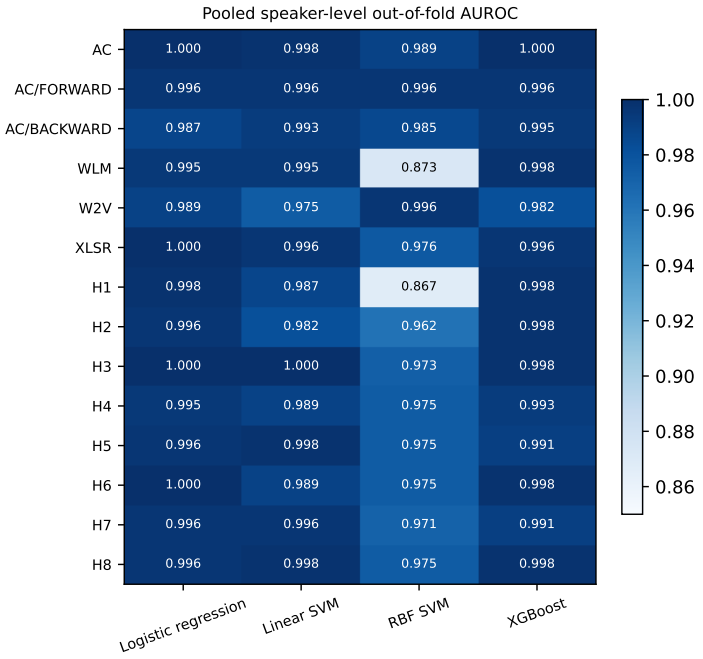


Figure 5: Pooled speaker-level AUROC for every representation and model (56 configurations).

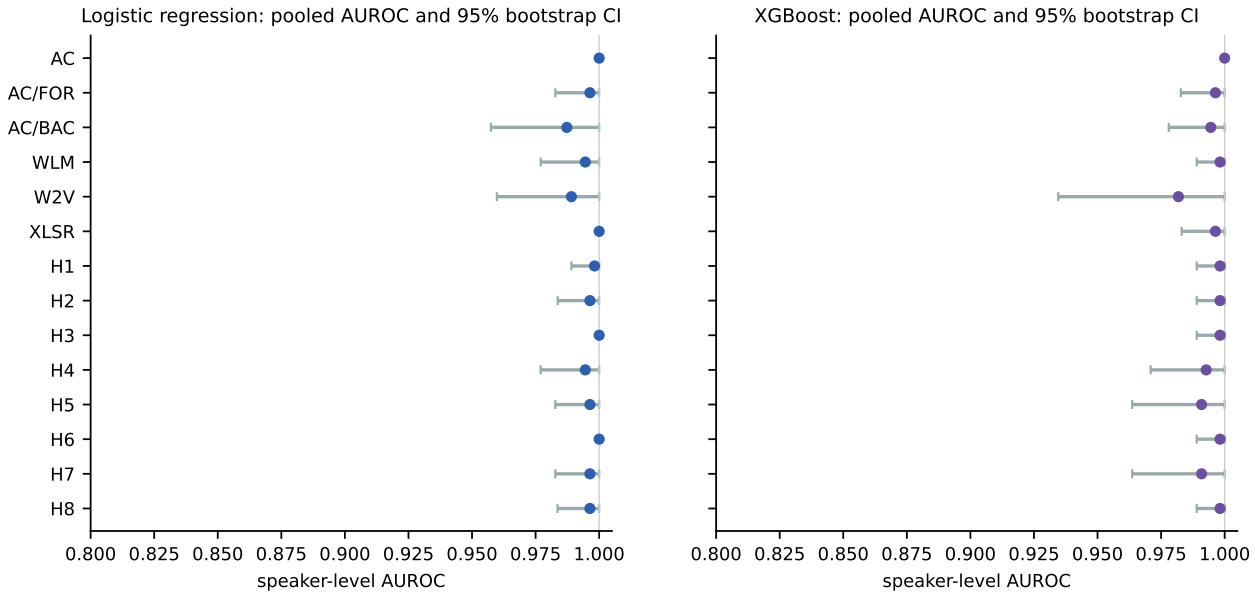


Figure 6: AUROC with 95% speaker-bootstrap intervals for logistic regression and XGBoost. Intervals collapse to a point when AUROC = 1 (percentile bootstrap at the boundary).

Table 4: All 56 configurations: pooled out-of-fold speaker-level metrics (47 speakers). Sensitivity and specificity with exact Clopper-Pearson 95% intervals; FN/FP = missed patients / false alarms at threshold 0.5.

Representation	Model	AUPRC	AUROC [bootstrap CI]	Sensitivity (exact CI)	Specificity (exact CI)	F1	Bal. acc.	FN/FP
Acoustic (40)	Logistic regression	1.000	1.000 [1.000, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
Acoustic (40)	Linear SVM	0.998	0.998 [0.989, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
Acoustic (40)	RBF SVM	0.991	0.989 [0.964, 1.000]	0.96 (0.80-1.00)	0.86 (0.65-0.97)	0.923	0.912	1/3
Acoustic (40)	XGBoost	1.000	1.000 [1.000, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
Acoustic (40) / forward	Logistic regression	0.997	0.996 [0.983, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
Acoustic (40) / forward	Linear SVM	0.997	0.996 [0.983, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
Acoustic (40) / forward	RBF SVM	0.997	0.996 [0.984, 1.000]	0.96 (0.80-1.00)	0.95 (0.77-1.00)	0.960	0.957	1/1
Acoustic (40) / forward	XGBoost	0.997	0.996 [0.983, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
Acoustic (40) / backward	Logistic regression	0.990	0.987 [0.957, 1.000]	0.96 (0.80-1.00)	0.91 (0.71-0.99)	0.941	0.935	1/2
Acoustic (40) / backward	Linear SVM	0.994	0.993 [0.971, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
Acoustic (40) / backward	RBF SVM	0.990	0.985 [0.947, 1.000]	0.96 (0.80-1.00)	0.86 (0.65-0.97)	0.923	0.912	1/3
Acoustic (40) / backward	XGBoost	0.996	0.995 [0.978, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
WavLM-large	Logistic regression	0.996	0.995 [0.977, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
WavLM-large	Linear SVM	0.996	0.995 [0.977, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
WavLM-large	RBF SVM	0.887	0.873 [0.752, 0.966]	0.96 (0.80-1.00)	0.77 (0.55-0.92)	0.889	0.866	1/5
WavLM-large	XGBoost	0.998	0.998 [0.989, 1.000]	0.96 (0.80-1.00)	0.95 (0.77-1.00)	0.960	0.957	1/1
wav2vec2-large-lv60	Logistic regression	0.992	0.989 [0.960, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
wav2vec2-large-lv60	Linear SVM	0.986	0.975 [0.913, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
wav2vec2-large-lv60	RBF SVM	0.997	0.996 [0.984, 1.000]	1.00 (0.86-1.00)	0.91 (0.71-0.99)	0.962	0.955	0/2
wav2vec2-large-lv60	XGBoost	0.989	0.982 [0.935, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
XLS-R 300M	Logistic regression	1.000	1.000 [1.000, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
XLS-R 300M	Linear SVM	0.997	0.996 [0.984, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
XLS-R 300M	RBF SVM	0.986	0.976 [0.919, 1.000]	0.96 (0.80-1.00)	0.91 (0.71-0.99)	0.941	0.935	1/2
XLS-R 300M	XGBoost	0.997	0.996 [0.983, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
WavLM + acoustic	Logistic regression	0.998	0.998 [0.989, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
WavLM + acoustic	Linear SVM	0.991	0.987 [0.955, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
WavLM + acoustic	RBF SVM	0.901	0.867 [0.749, 0.954]	0.96 (0.80-1.00)	0.59 (0.36-0.79)	0.828	0.775	1/9
WavLM + acoustic	XGBoost	0.998	0.998 [0.989, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
wav2vec2 + acoustic	Logistic regression	0.997	0.996 [0.984, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
wav2vec2 + acoustic	Linear SVM	0.989	0.982 [0.935, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
wav2vec2 + acoustic	RBF SVM	0.956	0.962 [0.891, 1.000]	1.00 (0.86-1.00)	0.73 (0.50-0.89)	0.893	0.864	0/6
wav2vec2 + acoustic	XGBoost	0.998	0.998 [0.989, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
XLS-R + acoustic	Logistic regression	1.000	1.000 [1.000, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
XLS-R + acoustic	Linear SVM	1.000	1.000 [1.000, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
XLS-R + acoustic	RBF SVM	0.985	0.973 [0.905, 1.000]	0.96 (0.80-1.00)	0.91 (0.71-0.99)	0.941	0.935	1/2
XLS-R + acoustic	XGBoost	0.998	0.998 [0.989, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
WavLM + wav2vec2	Logistic regression	0.996	0.995 [0.977, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
WavLM + wav2vec2	Linear SVM	0.992	0.989 [0.960, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
WavLM + wav2vec2	RBF SVM	0.986	0.975 [0.912, 1.000]	0.96 (0.80-1.00)	0.95 (0.77-1.00)	0.960	0.957	1/1
WavLM + wav2vec2	XGBoost	0.994	0.993 [0.971, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
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wav2vec2 + XLS-R	Logistic regression	1.000	1.000 [1.000, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
wav2vec2 + XLS-R	Linear SVM	0.992	0.989 [0.961, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
wav2vec2 + XLS-R	RBF SVM	0.986	0.975 [0.912, 1.000]	0.96 (0.80-1.00)	0.95 (0.77-1.00)	0.960	0.957	1/1
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three encoders	Logistic regression	0.997	0.996 [0.983, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
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three encoders	RBF SVM	0.984	0.971 [0.901, 1.000]	0.96 (0.80-1.00)	0.91 (0.71-0.99)	0.941	0.935	1/2
three encoders	XGBoost	0.993	0.991 [0.964, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
three encoders + acoustic	Logistic regression	0.997	0.996 [0.984, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
three encoders + acoustic	Linear SVM	0.998	0.998 [0.989, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0
three encoders + acoustic	RBF SVM	0.986	0.975 [0.912, 1.000]	0.96 (0.80-1.00)	0.91 (0.71-0.99)	0.941	0.935	1/2
three encoders + acoustic	XGBoost	0.998	0.998 [0.989, 1.000]	0.96 (0.80-1.00)	1.00 (0.85-1.00)	0.980	0.980	1/0

Table 5: Feature-combination leaderboard (median over the four models).

Combination	Dim.	Median AUROC	Min	Max	Median AUPRC	Fewest errors	Median errors
Acoustic (40)	40	0.999	0.989	1.000	0.999	1	1
WavLM-large	1024	0.995	0.873	0.998	0.996	1	1
wav2vec2-large-lv60	1024	0.985	0.975	0.996	0.990	1	1
XLS-R 300M	1024	0.996	0.976	1.000	0.997	1	1
WavLM + acoustic	1064	0.993	0.867	0.998	0.995	1	1
wav2vec2 + acoustic	1064	0.989	0.962	0.998	0.993	1	1
XLS-R + acoustic	1064	0.999	0.973	1.000	0.999	1	1
WavLM + wav2vec2	2048	0.991	0.975	0.995	0.993	1	1
WavLM + XLS-R	2048	0.994	0.975	0.998	0.995	1	1
wav2vec2 + XLS-R	2048	0.994	0.975	1.000	0.995	1	1
three encoders	3072	0.994	0.971	0.996	0.995	1	1
three encoders + acoustic	3112	0.997	0.975	0.998	0.998	1	1

Do combinations help? Over 76 paired comparisons of a hybrid with each of its components (same model, same folds) the median AUROC difference is 0.000 (range -0.122 to 0.102); the 95% interval excludes zero in 5 comparisons (4 up, 1 down). **Do non-linear models help?** Median AUROC difference against logistic regression: Linear SVM -0.001; RBF SVM -0.023; XGBoost -0.001. **Does feature selection help?** Ten selected acoustic features minus all 40: median -0.004, interval excluding zero in 0 of 8. In short, at this ceiling no combination, model or selection beats the simplest configurations; the differences are within one or two speakers.

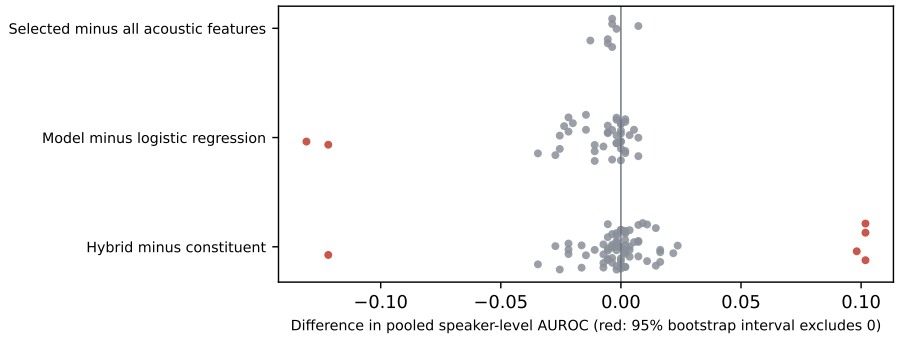


Figure 7: Paired AUROC differences for all pre-specified comparisons (red: interval excludes zero).

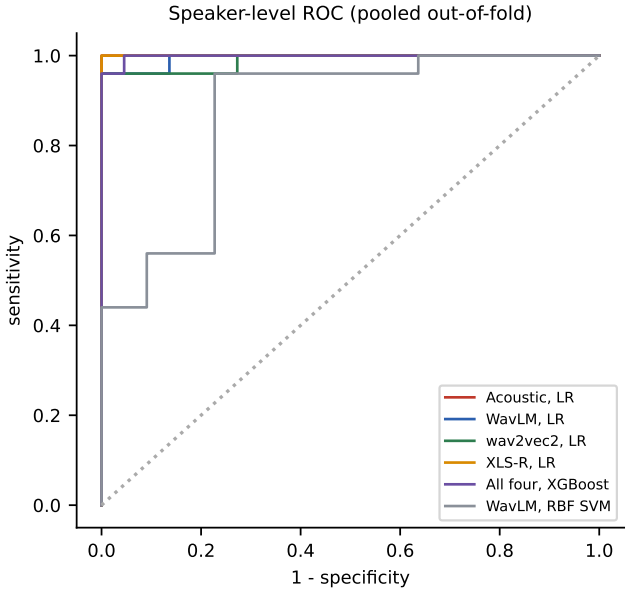


Figure 8: Pooled speaker-level ROC curves for six configurations.

Table 6: Fold-by-fold held-out results for eight representative configurations (9 or 10 test speakers per fold).

Configuration	Fold	Test speakers	AUPRC	AUROC	F1	FN/FP
AC LR	0	10	1.000	1.000	1.000	0/0
AC LR	1	10	1.000	1.000	1.000	0/0
AC LR	2	9	1.000	1.000	1.000	0/0
AC LR	3	9	1.000	1.000	0.889	1/0
AC LR	4	9	1.000	1.000	1.000	0/0
AC XGB	0	10	1.000	1.000	1.000	0/0
AC XGB	1	10	1.000	1.000	1.000	0/0
AC XGB	2	9	1.000	1.000	1.000	0/0
AC XGB	3	9	1.000	1.000	0.889	1/0
AC XGB	4	9	1.000	1.000	1.000	0/0
WLM LR	0	10	1.000	1.000	1.000	0/0
WLM LR	1	10	1.000	1.000	1.000	0/0
WLM LR	2	9	1.000	1.000	1.000	0/0
WLM LR	3	9	0.967	0.950	0.889	1/0
WLM LR	4	9	1.000	1.000	1.000	0/0
WLM XGB	0	10	1.000	1.000	1.000	0/0
WLM XGB	1	10	1.000	1.000	0.909	0/1
WLM XGB	2	9	1.000	1.000	1.000	0/0
WLM XGB	3	9	1.000	1.000	0.889	1/0
WLM XGB	4	9	1.000	1.000	1.000	0/0
W2V LR	0	10	1.000	1.000	1.000	0/0
W2V LR	1	10	1.000	1.000	1.000	0/0
W2V LR	2	9	1.000	1.000	1.000	0/0
W2V LR	3	9	0.911	0.800	0.889	1/0
W2V LR	4	9	1.000	1.000	1.000	0/0
XLSR LR	0	10	1.000	1.000	1.000	0/0
XLSR LR	1	10	1.000	1.000	1.000	0/0
XLSR LR	2	9	1.000	1.000	1.000	0/0
XLSR LR	3	9	1.000	1.000	0.889	1/0
XLSR LR	4	9	1.000	1.000	1.000	0/0
H3 LR	0	10	1.000	1.000	1.000	0/0
H3 LR	1	10	1.000	1.000	1.000	0/0
H3 LR	2	9	1.000	1.000	1.000	0/0
H3 LR	3	9	1.000	1.000	0.889	1/0
H3 LR	4	9	1.000	1.000	1.000	0/0
H8 XGB	0	10	1.000	1.000	1.000	0/0
H8 XGB	1	10	1.000	1.000	1.000	0/0
H8 XGB	2	9	1.000	1.000	1.000	0/0
H8 XGB	3	9	1.000	1.000	0.889	1/0
H8 XGB	4	9	1.000	1.000	1.000	0/0

Table 7: Training versus inner-validation versus held-out scores (means over the five outer folds; training scores from a diagnostic refit). Perfect training fit with lower but close validation and held-out scores: no sign that tuning overfitted the held-out folds.

Configuration	Training AUPRC / AUROC	Inner-validation AUPRC	Held-out fold AUPRC
AC LR	1.000 / 1.000	0.998 (SD 0.003)	1.000 (SD 0.000)
WLM LR	1.000 / 1.000	0.999 (SD 0.001)	0.993 (SD 0.015)
W2V LR	1.000 / 1.000	0.988 (SD 0.008)	0.982 (SD 0.040)
XLSR LR	1.000 / 1.000	0.999 (SD 0.001)	1.000 (SD 0.000)
H3 LR	1.000 / 1.000	0.999 (SD 0.001)	1.000 (SD 0.000)
WLM XGB	1.000 / 1.000	0.997 (SD 0.003)	1.000 (SD 0.000)

Table 8: Recording-level versus speaker-level discrimination: averaging a speaker's recordings raises AUROC in almost every configuration.

Representation	Model	Recording AUROC	Recording AUPRC	Speaker AUROC	Speaker AUPRC
Acoustic (40)	Logistic regression	0.969	0.973	1.000	1.000
Acoustic (40)	Linear SVM	0.964	0.970	0.998	0.998
Acoustic (40)	RBF SVM	0.977	0.983	0.989	0.991
Acoustic (40)	XGBoost	0.975	0.985	1.000	1.000
Acoustic (40) / forward	Logistic regression	0.954	0.970	0.996	0.997
Acoustic (40) / forward	Linear SVM	0.955	0.971	0.996	0.997
Acoustic (40) / forward	RBF SVM	0.972	0.980	0.996	0.997
Acoustic (40) / forward	XGBoost	0.950	0.966	0.996	0.997
Acoustic (40) / backward	Logistic regression	0.902	0.921	0.987	0.990
Acoustic (40) / backward	Linear SVM	0.909	0.930	0.993	0.994
Acoustic (40) / backward	RBF SVM	0.948	0.965	0.985	0.990
Acoustic (40) / backward	XGBoost	0.961	0.972	0.995	0.996
WavLM-large	Logistic regression	0.959	0.963	0.995	0.996
WavLM-large	Linear SVM	0.980	0.990	0.995	0.996
WavLM-large	RBF SVM	0.869	0.888	0.873	0.887
WavLM-large	XGBoost	0.974	0.979	0.998	0.998
wav2vec2-large-lv60	Logistic regression	0.978	0.985	0.989	0.992
wav2vec2-large-lv60	Linear SVM	0.969	0.983	0.975	0.986
wav2vec2-large-lv60	RBF SVM	0.973	0.979	0.996	0.997
wav2vec2-large-lv60	XGBoost	0.957	0.971	0.982	0.989
XLS-R 300M	Logistic regression	0.985	0.991	1.000	1.000
XLS-R 300M	Linear SVM	0.980	0.989	0.996	0.997
XLS-R 300M	RBF SVM	0.963	0.978	0.976	0.986
XLS-R 300M	XGBoost	0.978	0.983	0.996	0.997
WavLM + acoustic	Logistic regression	0.984	0.991	0.998	0.998
WavLM + acoustic	Linear SVM	0.978	0.989	0.987	0.991
WavLM + acoustic	RBF SVM	0.849	0.886	0.867	0.901
WavLM + acoustic	XGBoost	0.976	0.987	0.998	0.998
wav2vec2 + acoustic	Logistic regression	0.981	0.988	0.996	0.997
wav2vec2 + acoustic	Linear SVM	0.972	0.986	0.982	0.989

Representation	Model	Recording AUROC	Recording AUPRC	Speaker AUROC	Speaker AUPRC
wav2vec2 + acoustic	RBF SVM	0.929	0.936	0.962	0.956
wav2vec2 + acoustic	XGBoost	0.977	0.986	0.998	0.998
XLS-R + acoustic	Logistic regression	0.981	0.990	1.000	1.000
XLS-R + acoustic	Linear SVM	0.981	0.991	1.000	1.000
XLS-R + acoustic	RBF SVM	0.963	0.978	0.973	0.985
XLS-R + acoustic	XGBoost	0.978	0.987	0.998	0.998
WavLM + wav2vec2	Logistic regression	0.975	0.981	0.995	0.996
WavLM + wav2vec2	Linear SVM	0.978	0.989	0.989	0.992
WavLM + wav2vec2	RBF SVM	0.958	0.971	0.975	0.986
WavLM + wav2vec2	XGBoost	0.968	0.973	0.993	0.994
WavLM + XLS-R	Logistic regression	0.979	0.983	0.996	0.997
WavLM + XLS-R	Linear SVM	0.982	0.991	0.998	0.998
WavLM + XLS-R	RBF SVM	0.955	0.970	0.975	0.986
WavLM + XLS-R	XGBoost	0.976	0.981	0.991	0.993
wav2vec2 + XLS-R	Logistic regression	0.981	0.990	1.000	1.000
wav2vec2 + XLS-R	Linear SVM	0.979	0.990	0.989	0.992
wav2vec2 + XLS-R	RBF SVM	0.964	0.978	0.975	0.986
wav2vec2 + XLS-R	XGBoost	0.981	0.986	0.998	0.998
three encoders	Logistic regression	0.979	0.984	0.996	0.997
three encoders	Linear SVM	0.981	0.990	0.996	0.997
three encoders	RBF SVM	0.956	0.973	0.971	0.984
three encoders	XGBoost	0.973	0.980	0.991	0.993
three encoders + acoustic	Logistic regression	0.979	0.989	0.996	0.997
three encoders + acoustic	Linear SVM	0.980	0.990	0.998	0.998
three encoders + acoustic	RBF SVM	0.967	0.980	0.975	0.986
three encoders + acoustic	XGBoost	0.980	0.989	0.998	0.998

8. Every acoustic feature

Speaker-level univariate analysis of all 40 features (descriptive, whole cohort, not used for any model decision). 12 features separate the groups with direction-free AUROC of at least 0.90 on their own; all of them are spectral-shape features (zero-crossing rate, spectral centroid, roll-off), and **they are lower in PD**. The effect remains when only 16-kHz recordings are used, so it is not only a resampling artefact. But the share of variance explained by the recording date **within one class** is large for many features (within-control eta-squared up to 0.95), i.e. the same features also vary strongly between sessions of healthy people.

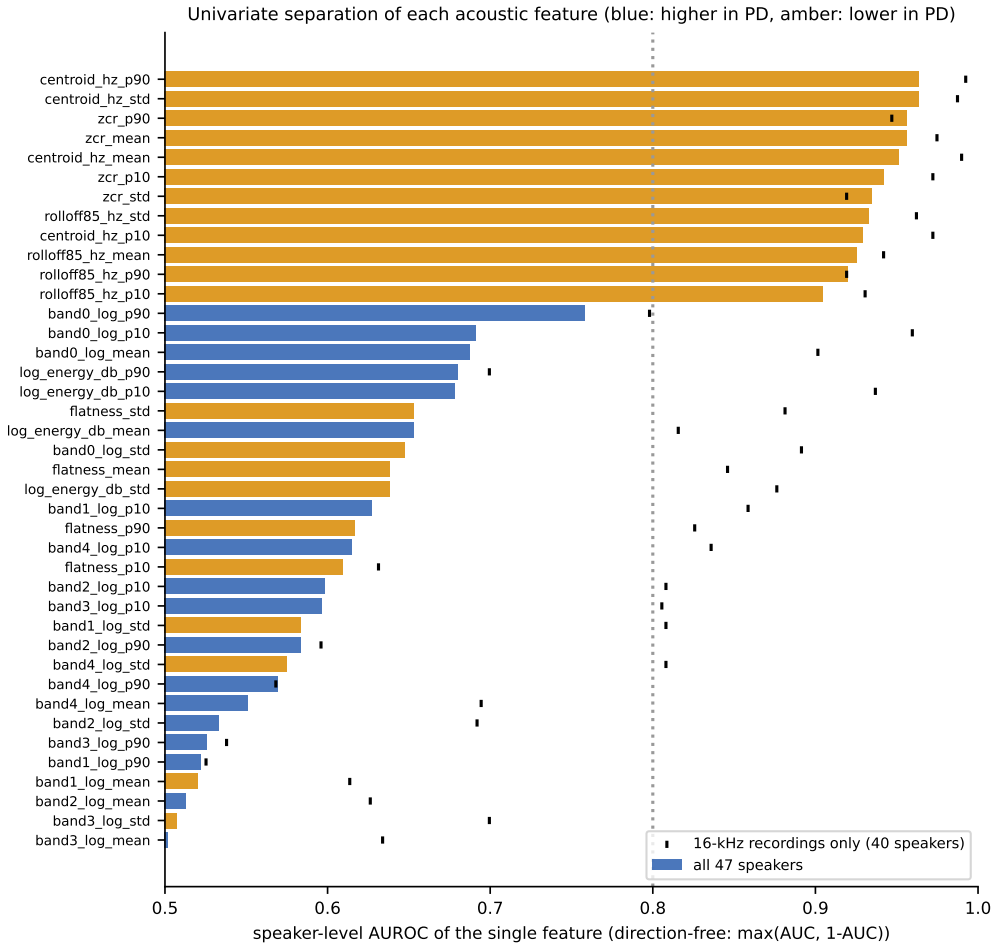


Figure 9: Univariate speaker-level separation of each of the 40 acoustic features (bars) and the same analysis on 16-kHz recordings only (ticks).

Table 9: Feature dictionary and univariate statistics (speaker means; AUROC $>$ 0.5 means higher in PD). eta2: share of variance explained by class, or by recording date within PD / within controls.

Feature	Direction	Mean PD	Mean ctrl	AUROC	AUROC 16 kHz	Cohen d	eta2 class	eta2 date|PD	eta2 date|ctrl
centroid_hz_p90	PD lower	615.997	1197.133	0.036	0.008	-2.19	0.56	0.43	0.87
centroid_hz_std	PD lower	265.736	506.689	0.036	0.013	-1.56	0.39	0.38	0.95
zcr_p90	PD lower	0.117	0.199	0.044	0.053	-2.51	0.62	0.37	0.32
zcr_mean	PD lower	0.066	0.111	0.044	0.025	-2.74	0.66	0.34	0.37
centroid_hz_mean	PD lower	405.866	714.395	0.049	0.010	-2.76	0.66	0.38	0.22
zcr_p10	PD lower	0.030	0.058	0.058	0.028	-2.31	0.58	0.43	0.50
zcr_std	PD lower	0.045	0.076	0.065	0.081	-2.06	0.52	0.13	0.63
rolloff85_hz_std	PD lower	529.889	894.172	0.067	0.038	-1.44	0.35	0.33	0.91
centroid_hz_p10	PD lower	225.041	421.568	0.071	0.028	-2.14	0.54	0.45	0.38
rolloff85_hz_mean	PD lower	637.116	1038.761	0.075	0.058	-2.08	0.53	0.40	0.31
rolloff85_hz_p90	PD lower	1132.149	2016.296	0.080	0.081	-1.75	0.44	0.51	0.77
rolloff85_hz_p10	PD lower	300.355	517.465	0.095	0.069	-1.79	0.45	0.43	0.34
band0_log_p90	PD higher	2.077	1.605	0.758	0.798	0.94	0.19	0.24	0.61
band0_log_p10	PD higher	-2.170	-2.102	0.691	0.960	-0.03	0.00	0.83	0.15
band0_log_mean	PD higher	0.426	-0.058	0.687	0.902	0.65	0.10	0.64	0.61
log_energy_db_p90	PD higher	-23.201	-26.289	0.680	0.699	0.62	0.09	0.12	0.53
log_energy_db_p10	PD higher	-62.285	-62.708	0.678	0.937	0.02	0.00	0.81	0.08
flatness_std	PD lower	0.076	0.055	0.347	0.119	0.29	0.02	0.71	0.89
log_energy_db_mean	PD higher	-38.960	-42.448	0.653	0.816	0.51	0.06	0.54	0.54
band0_log_std	PD lower	2.141	1.559	0.353	0.109	0.45	0.05	0.70	0.74
log_energy_db_std	PD lower	18.972	15.062	0.362	0.124	0.40	0.04	0.69	0.64
flatness_mean	PD lower	0.032	0.028	0.362	0.154	0.14	0.00	0.76	0.77
band1_log_p10	PD higher	-3.384	-2.652	0.627	0.859	-0.32	0.03	0.81	0.13
flatness_p90	PD lower	0.118	0.088	0.384	0.174	0.25	0.02	0.87	0.84
band4_log_p10	PD higher	-4.764	-4.057	0.615	0.836	-0.35	0.03	0.83	0.42
flatness_p10	PD lower	0.000	0.001	0.391	0.369	-0.25	0.02	0.28	0.35
band2_log_p10	PD higher	-4.160	-3.393	0.598	0.808	-0.35	0.03	0.82	0.33
band3_log_p10	PD higher	-4.437	-3.593	0.596	0.806	-0.40	0.04	0.81	0.21
band2_log_p90	PD higher	-0.026	-0.245	0.584	0.596	0.29	0.02	0.16	0.56
band1_log_std	PD lower	2.199	1.547	0.416	0.192	0.57	0.08	0.68	0.67
band4_log_std	PD lower	1.706	1.047	0.425	0.192	0.60	0.09	0.70	0.41
band4_log_p90	PD higher	-1.330	-1.474	0.569	0.568	0.21	0.01	0.01	0.27
band4_log_mean	PD higher	-2.764	-2.838	0.551	0.694	0.10	0.00	0.50	0.45
band2_log_std	PD higher	1.981	1.260	0.533	0.308	0.66	0.10	0.69	0.75
band3_log_p90	PD higher	-0.610	-0.620	0.525	0.538	0.01	0.00	0.10	0.42
band1_log_p90	PD higher	1.175	1.142	0.522	0.525	0.05	0.00	0.09	0.34
band1_log_mean	PD lower	-0.778	-0.636	0.480	0.614	-0.17	0.01	0.52	0.33
band2_log_mean	PD higher	-1.808	-1.803	0.513	0.626	-0.01	0.00	0.53	0.59
band3_log_std	PD lower	1.867	1.180	0.493	0.301	0.64	0.10	0.70	0.61
band3_log_mean	PD higher	-2.227	-2.120	0.502	0.634	-0.13	0.00	0.50	0.48

Table 10: Feature families (mean and best direction-free AUROC of their four statistics; mean within-class date effect).

Family	Mean AUROC	Best AUROC	Mean eta2 date|PD	Mean eta2 date|ctrl
centroid (Hz)	0.952	0.964	0.41	0.61
zcr	0.947	0.956	0.32	0.45
rolloff85 (Hz)	0.921	0.933	0.42	0.58
band0 (log)	0.696	0.758	0.60	0.53
log_energy_db	0.662	0.680	0.54	0.45
flatness	0.629	0.653	0.66	0.71
band4 (log)	0.577	0.615	0.51	0.39
band1 (log)	0.563	0.627	0.52	0.37
band2 (log)	0.557	0.598	0.55	0.56
band3 (log)	0.533	0.596	0.53	0.43

Redundancy: 46 feature pairs have an absolute Spearman correlation of at least 0.9 at speaker level (for example the mean and the percentiles of the same frame feature), so single-feature importance is diluted and family-level analysis is more informative.

9. Feature families, statistics and feature selection

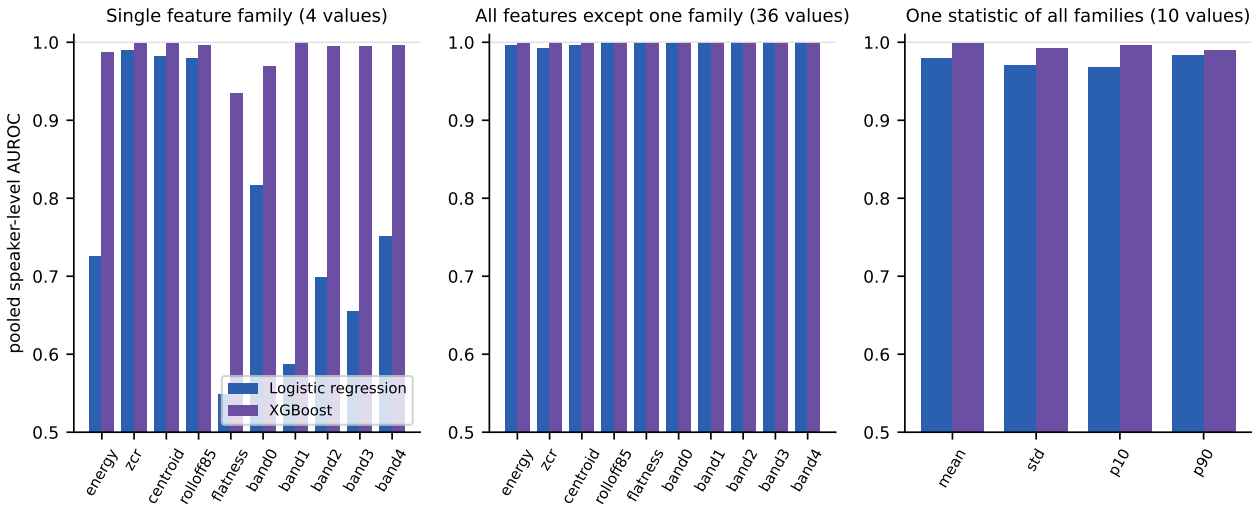


Figure 10: Exploratory ablations with the same nested protocol: one family alone, all but one family, and one statistic of all families.

Table 11: Ablation results (66 configurations, exploratory amendment A1, same folds and nested tuning).

Feature subset	Model	Features	AUROC	AUPRC	F1	FN/FP
drop: log_energy_db	Logistic regression	36	0.996	0.997	0.980	1/0
drop: log_energy_db	XGBoost	36	0.998	0.998	0.980	1/0
only: log_energy_db	Logistic regression	4	0.725	0.727	0.741	5/9
only: log_energy_db	XGBoost	4	0.987	0.990	0.939	2/1
drop: zcr	Logistic regression	36	0.993	0.994	0.980	1/0
drop: zcr	XGBoost	36	1.000	1.000	0.980	1/0
only: zcr	Logistic regression	4	0.989	0.992	0.980	1/0
only: zcr	XGBoost	4	0.998	0.998	0.980	1/0
drop: centroid	Logistic regression	36	0.996	0.997	0.960	1/1
drop: centroid	XGBoost	36	0.998	0.998	0.980	1/0
only: centroid	Logistic regression	4	0.982	0.989	0.958	2/0
only: centroid	XGBoost	4	0.998	0.998	0.958	2/0
drop: rolloff85	Logistic regression	36	1.000	1.000	0.980	1/0
drop: rolloff85	XGBoost	36	1.000	1.000	0.980	1/0
only: rolloff85	Logistic regression	4	0.980	0.985	0.939	2/1
only: rolloff85	XGBoost	4	0.996	0.997	0.980	1/0
drop: flatness	Logistic regression	36	1.000	1.000	0.980	1/0
drop: flatness	XGBoost	36	1.000	1.000	0.980	1/0
only: flatness	Logistic regression	4	0.549	0.654	0.400	18/3
only: flatness	XGBoost	4	0.935	0.959	0.898	3/2
drop: band0	Logistic regression	36	0.998	0.998	0.960	1/1
drop: band0	XGBoost	36	1.000	1.000	1.000	0/0

Feature subset	Model	Features	AUROC	AUPRC	F1	FN/FP
only: band0	Logistic regression	4	0.816	0.800	0.750	7/5
only: band0	XGBoost	4	0.969	0.980	0.939	2/1
drop: band1	Logistic regression	36	1.000	1.000	0.980	1/0
drop: band1	XGBoost	36	1.000	1.000	0.980	1/0
only: band1	Logistic regression	4	0.587	0.704	0.545	13/7
only: band1	XGBoost	4	1.000	1.000	0.980	1/0
drop: band2	Logistic regression	36	1.000	1.000	0.980	1/0
drop: band2	XGBoost	36	1.000	1.000	0.980	1/0
only: band2	Logistic regression	4	0.698	0.742	0.591	12/6
only: band2	XGBoost	4	0.995	0.996	0.980	1/0
drop: band3	Logistic regression	36	1.000	1.000	0.980	1/0
drop: band3	XGBoost	36	1.000	1.000	0.980	1/0
only: band3	Logistic regression	4	0.655	0.730	0.524	14/6
only: band3	XGBoost	4	0.995	0.996	0.960	1/1
drop: band4	Logistic regression	36	1.000	1.000	1.000	0/0
drop: band4	XGBoost	36	1.000	1.000	0.980	1/0
only: band4	Logistic regression	4	0.751	0.762	0.591	12/6
only: band4	XGBoost	4	0.996	0.997	0.980	1/0
only statistic: mean	Logistic regression	10	0.980	0.986	0.917	3/1
only statistic: mean	XGBoost	10	0.998	0.998	0.958	2/0
drop statistic: mean	Logistic regression	30	1.000	1.000	1.000	0/0
drop statistic: mean	XGBoost	30	0.998	0.998	0.980	1/0
only statistic: std	Logistic regression	10	0.971	0.969	0.941	1/2
only statistic: std	XGBoost	10	0.993	0.994	0.980	1/0
drop statistic: std	Logistic regression	30	0.989	0.991	0.960	1/1
drop statistic: std	XGBoost	30	0.996	0.997	0.958	2/0
only statistic: p10	Logistic regression	10	0.967	0.969	0.920	2/2
only statistic: p10	XGBoost	10	0.996	0.997	0.960	1/1
drop statistic: p10	Logistic regression	30	0.993	0.994	0.958	2/0
drop statistic: p10	XGBoost	30	0.996	0.997	0.980	1/0
only statistic: p90	Logistic regression	10	0.984	0.989	0.980	1/0
only statistic: p90	XGBoost	10	0.989	0.992	0.980	1/0
drop statistic: p90	Logistic regression	30	0.991	0.993	0.980	1/0
drop statistic: p90	XGBoost	30	1.000	1.000	0.980	1/0
only: spectral_shape	Logistic regression	12	0.985	0.990	0.980	1/0
only: spectral_shape	XGBoost	12	0.998	0.998	0.958	2/0
only: band_energies	Logistic regression	20	0.993	0.994	0.923	1/3
only: band_energies	XGBoost	20	0.993	0.994	0.980	1/0
only: zcr_and_shape	Logistic regression	16	0.991	0.993	0.980	1/0
only: zcr_and_shape	XGBoost	16	0.996	0.997	0.980	1/0

Reading: no family is necessary (dropping any single family leaves AUROC $\geq$ 0.993), and several are sufficient on their own with XGBoost (for example the 500-1000 Hz band: AUROC 1.000). With logistic regression, the four zero-crossing statistics alone reach 0.989 and energy alone only 0.725, so the linear signal is mostly spectral shape. Any single statistic over all families (10 values) reaches at least 0.967. **Sequential selection (frozen FINAL_V1 conditions).** Forward and backward selection of exactly 10 of the 40 features, repeated inside every training partition with a logistic-regression selector (selection is identical for the four final models). Forward: AUROC 0.996-0.996 across models; backward: 0.985-0.995; all 40: 0.989-1.000. Stability is moderate: the table lists how often each feature was selected over the five outer folds.

Table 12: Selected features and selection frequency over the five outer folds.

Method	Feature	Selected
forward	log_energy_db_mean	4/5
forward	log_energy_db_p10	4/5
forward	zcr_mean	4/5
forward	zcr_p10	4/5
forward	centroid_hz_std	4/5
forward	band2_log_p90	4/5
forward	log_energy_db_p90	4/5
forward	centroid_hz_p90	4/5
forward	log_energy_db_std	3/5
forward	zcr_p90	3/5
forward	zcr_std	3/5
forward	centroid_hz_p10	2/5
forward	rolloff85_hz_mean	2/5
forward	band1_log_std	1/5
forward	flatness_p10	1/5
forward	band1_log_p90	1/5
forward	band4_log_mean	1/5
forward	centroid_hz_mean	1/5
backward	band4_log_mean	5/5
backward	band4_log_p10	5/5
backward	band4_log_p90	5/5
backward	band3_log_p10	4/5
backward	band4_log_std	4/5
backward	rolloff85_hz_std	3/5
backward	rolloff85_hz_p90	3/5
backward	band3_log_std	3/5
backward	band3_log_p90	3/5
backward	band2_log_p90	3/5
backward	centroid_hz_p10	2/5
backward	centroid_hz_p90	2/5
backward	rolloff85_hz_mean	2/5
backward	zcr_p10	2/5
backward	log_energy_db_p90	1/5
backward	band0_log_p90	1/5
backward	centroid_hz_std	1/5
backward	band1_log_mean	1/5

10. Feature importance

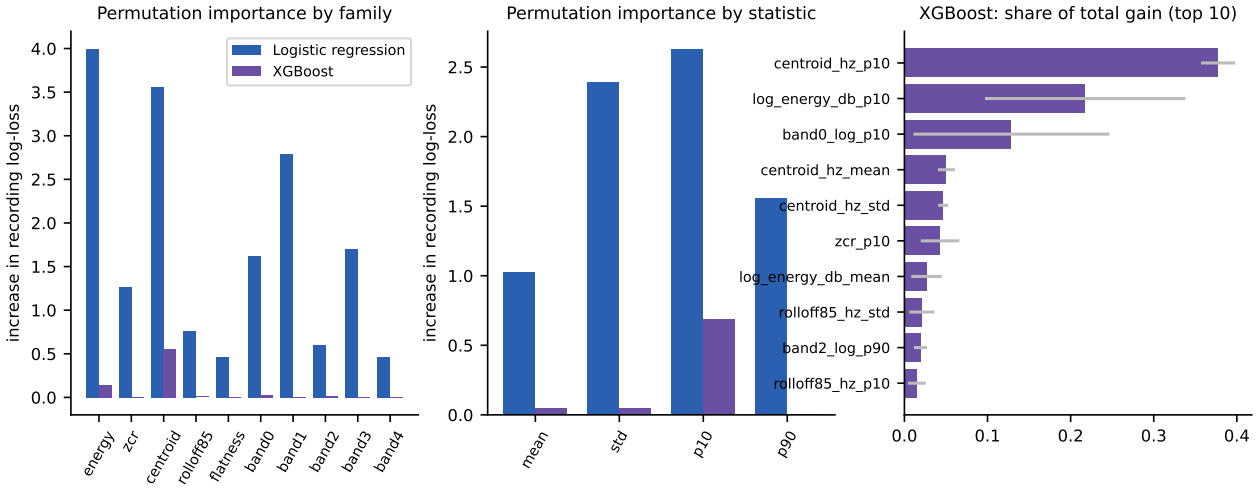


Figure 11: Acoustic models: permutation importance (increase in recording-level log-loss when a family or a statistic is shuffled within the outer-test folds) and XGBoost gain shares.

Logistic regression relies on many correlated features (top by permutation: band3_log_std, log_energy_db_std, log_energy_db_mean, zcr_p10, centroid_hz_std); its coefficients change sign across folds for some energy features, a sign of collinearity. **XGBoost** concentrates on **10th-percentile statistics**: centroid_hz_p10, log_energy_db_p10, band0_log_p10 alone carry 72% of the total gain, and shuffling the p10 statistic of all families costs more than any family. The 10th percentile of a frame feature describes the quietest frames, i.e. pauses and background, not the voice.

Table 13: Logistic regression on acoustic features: top 15 features by permutation importance (mean of 20 shuffles) and the mean standardised coefficient over the five outer folds.

Feature	Log-loss increase	AUROC drop	Mean coefficient	Sign stable
band3_log_std	2.258	0.095	18.66	yes
log_energy_db_std	2.044	0.057	-22.58	no
log_energy_db_mean	1.779	0.046	12.63	no
zcr_p10	1.679	0.056	-9.01	yes
centroid_hz_std	1.536	0.053	-7.54	yes
band1_log_mean	1.395	0.017	-8.85	yes
band3_log_p90	1.296	0.037	-8.85	yes
band1_log_std	1.280	0.021	10.89	yes
band0_log_mean	1.077	0.013	-9.22	yes
log_energy_db_p90	0.995	0.022	7.56	yes
centroid_hz_mean	0.851	0.007	-5.88	no
flatness_std	0.811	0.019	-9.40	no
band3_log_mean	0.767	0.004	7.64	yes
band4_log_std	0.716	0.005	8.09	no
band0_log_p90	0.568	0.008	-5.92	yes

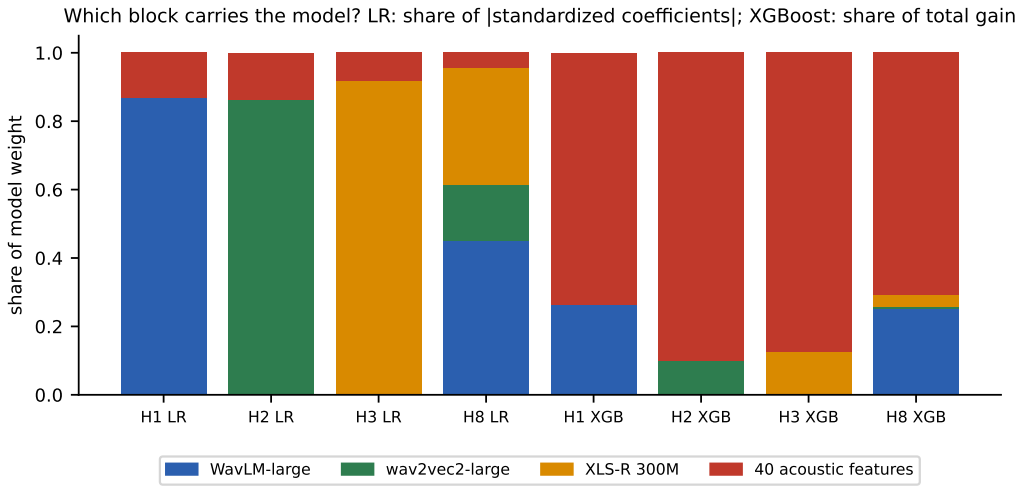


Figure 12: Hybrid models: which representation block carries the weight.

In hybrids the two model families disagree about what matters. Logistic regression puts 86%-92% of its coefficient mass on the encoder block, whereas XGBoost puts 74%-90% of its gain on the 40 acoustic features. When all three encoders are combined, logistic regression weights WavLM most (50%), then XLS-R (35%), then wav2vec2 (15%). Shuffling the wav2vec2 block in the three-encoder models changes nothing, i.e. it is redundant given the other two.

Table 14: Block permutation importance and weight share in hybrids (LR: coefficient mass; XGBoost: gain).

Model	Block	AUROC drop	Log-loss increase	Weight share
H1 LR	WLM	0.186	1.329	87%
H1 LR	AC	0.011	0.186	13%
H1 XGB	WLM	0.002	0.124	26%
H1 XGB	AC	0.127	0.577	74%
H2 LR	W2V	0.055	1.144	86%
H2 LR	AC	0.032	0.732	14%
H2 XGB	W2V	-0.000	0.035	10%
H2 XGB	AC	0.385	1.073	90%
H3 LR	XLSR	0.211	1.085	92%
H3 LR	AC	0.005	0.142	8%
H3 XGB	XLSR	0.000	0.054	13%
H3 XGB	AC	0.286	0.705	87%
H7 LR	WLM	0.081	0.472	50%
H7 LR	W2V	-0.000	0.015	15%
H7 LR	XLSR	0.002	0.226	35%
H7 XGB	WLM	0.102	0.539	73%
H7 XGB	W2V	-0.000	0.017	6%
H7 XGB	XLSR	0.003	0.101	21%
H8 LR	WLM	0.020	0.433	45%
H8 LR	W2V	-0.002	-0.001	16%
H8 LR	XLSR	0.018	0.245	34%
H8 LR	AC	0.000	0.041	4%
H8 XGB	WLM	0.001	0.111	25%
H8 XGB	W2V	-0.000	0.001	1%
H8 XGB	XLSR	-0.000	0.020	3%
H8 XGB	AC	0.100	0.583	71%

11. Hyperparameter tuning

Table 15: Frozen search spaces (inner objective: pooled speaker-level AUPRC over three speaker-disjoint inner folds; ties to the first configuration).

Model	Configs	Grid
Logistic regression	20	C {0.01,0.1,1,10,100} x penalty {L1,L2} x class weight {none, balanced}
Linear SVM	10	C {0.01,0.1,1,10,100} x class weight
RBF SVM	24	C {0.1,1,10,100} x gamma = m/d, m {0.1,1,10} x class weight
XGBoost	24	24 fixed of 256: learning rate {0.03,0.1}, depth {2,4}, trees {100,300}, min child weight {1,5}, subsample {0.7,1}, colsample {0.3,1}, lambda {1,10}, pos. weight {1, balanced}

Finding 1 - the inner objective saturates. In most runs several configurations tie at the best inner score (Logistic regression 64 of 70 runs (median 12 tied); Linear SVM 63 of 70 runs (median 10 tied); RBF SVM 65 of 70 runs (median 6 tied); XGBoost 59 of 70 runs (median 9.5 tied)). Selection then falls back to the order of the grid, so 'tuning' is largely tie-breaking: logistic regression most often ends at C = 0.1 or 0.01 with L2, the linear SVM at C = 0.01, the RBF SVM at C = 1 with a narrow kernel (m = 10), XGBoost at the first configuration of its list.

Table 16: Most frequently selected configurations over the 70 runs of each model family.

Model	Configuration	Runs
Logistic regression	C=0.1, class_weight=None, penalty=l2	15 of 70
Logistic regression	C=0.01, class_weight=None, penalty=l2	14 of 70
Logistic regression	C=0.01, class_weight=None, penalty=l1	8 of 70
Logistic regression	C=1, class_weight=None, penalty=l1	6 of 70
Logistic regression	C=10, class_weight=None, penalty=l2	5 of 70
Linear SVM	C=0.01, class_weight=None	38 of 70
Linear SVM	C=0.1, class_weight=None	15 of 70
Linear SVM	C=1, class_weight=None	6 of 70
Linear SVM	C=10, class_weight=None	3 of 70
Linear SVM	C=0.01, class_weight=balanced	3 of 70
RBF SVM	C=1, class_weight=None, gamma_mult=10	26 of 70
RBF SVM	C=1, class_weight=None, gamma_mult=1	9 of 70
RBF SVM	C=0.1, class_weight=None, gamma_mult=10	7 of 70
RBF SVM	C=10, class_weight=None, gamma_mult=10	6 of 70
RBF SVM	C=0.1, class_weight=None, gamma_mult=1	5 of 70
XGBoost	colsample_bytree=1.0, learning_rate=0.03, max_depth=2, min_child_weight=1, n_estimators=100, reg_lambda=10, scale_pos_weight=1, subsample=0.7	28 of 70
XGBoost	colsample_bytree=1.0, learning_rate=0.03, max_depth=4, min_child_weight=1, n_estimators=100, reg_lambda=1, scale_pos_weight=1, subsample=0.7	8 of 70
XGBoost	colsample_bytree=0.3, learning_rate=0.03, max_depth=4, min_child_weight=1, n_estimators=300, reg_lambda=1, scale_pos_weight=1, subsample=0.7	6 of 70
XGBoost	colsample_bytree=1.0, learning_rate=0.1, max_depth=2, min_child_weight=1, n_estimators=300, reg_lambda=10, scale_pos_weight=balanced, subsample=0.7	5 of 70
XGBoost	colsample_bytree=0.3, learning_rate=0.1, max_depth=2, min_child_weight=1, n_estimators=100, reg_lambda=1, scale_pos_weight=1, subsample=1.0	4 of 70

Finding 2 - when tuning goes wrong it is the RBF SVM. Refitting every configuration on the outer folds (post hoc, descriptive) shows that most configurations of the linear models perform alike, while the RBF SVM spans a wide range (for wav2vec2 from 0.562 to 0.996). The largest loss from the nested choice is WavLM-large with RBF SVM: nested choice 0.873 against a grid median of 0.996. The RBF SVM's poor scores are therefore a tuning artefact of a saturated objective (plus an uncalibrated score at the fixed threshold), not evidence against the model family.

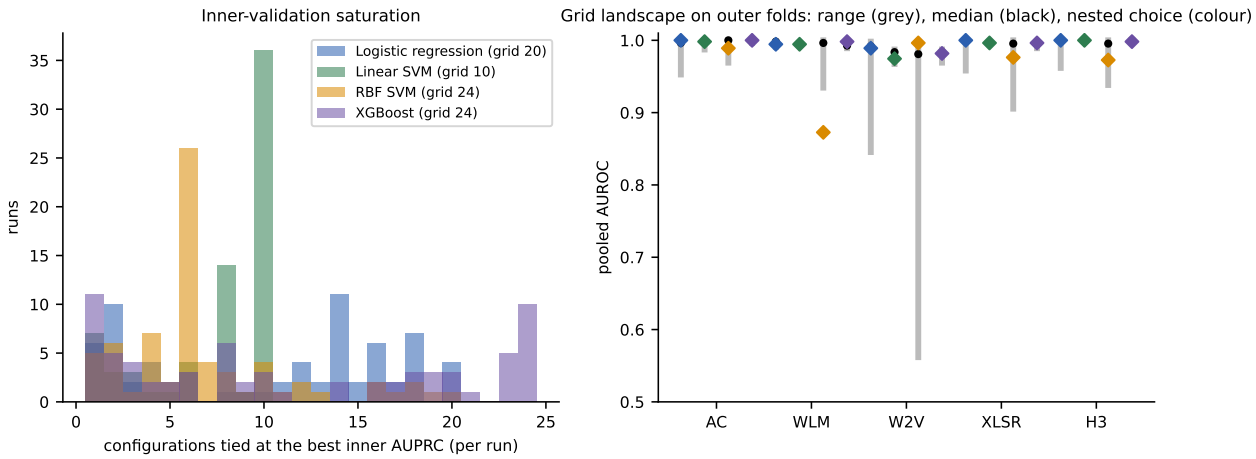


Figure 13: Left: number of configurations tied at the best inner score per run. Right: outer-fold AUROC of every grid configuration (range, median) and the nested choice.

Table 17: Tuning landscape: pooled AUROC of every configuration refitted on the outer folds (descriptive) versus the nested selection.

Representation	Model	Worst config	Median config	Best config	Nested choice
Acoustic (40)	Logistic regression	0.953	0.996	1.000	1.000
Acoustic (40)	Linear SVM	0.987	0.996	1.000	0.998
Acoustic (40)	RBF SVM	0.969	1.000	1.000	0.989
Acoustic (40)	XGBoost	0.996	1.000	1.000	1.000
XLS-R + acoustic	Logistic regression	0.962	1.000	1.000	1.000
XLS-R + acoustic	Linear SVM	1.000	1.000	1.000	1.000
XLS-R + acoustic	RBF SVM	0.938	0.995	1.000	0.973
XLS-R + acoustic	XGBoost	0.996	1.000	1.000	0.998
wav2vec2-large-lv60	Logistic regression	0.845	0.992	0.998	0.989
wav2vec2-large-lv60	Linear SVM	0.967	0.984	0.987	0.975
wav2vec2-large-lv60	RBF SVM	0.562	0.981	0.996	0.996

Representation	Model	Worst config	Median config	Best config	Nested choice
wav2vec2-large-lv60	XGBoost	0.969	0.984	0.987	0.982
WavLM-large	Logistic regression	0.991	0.998	0.998	0.995
WavLM-large	Linear SVM	0.995	0.995	0.995	0.995
WavLM-large	RBF SVM	0.935	0.996	1.000	0.873
WavLM-large	XGBoost	0.989	0.993	0.998	0.998
XLS-R 300M	Logistic regression	0.958	1.000	1.000	1.000
XLS-R 300M	Linear SVM	0.995	0.996	0.996	0.996
XLS-R 300M	RBF SVM	0.905	0.995	1.000	0.976
XLS-R 300M	XGBoost	0.989	0.995	0.995	0.996

Implication for product development. With few speakers per fold, classical grid search cannot discriminate between configurations. A product model should use simple, strongly regularised defaults, report sensitivity to them, and spend data on external validation rather than on tuning.

12. Which speech tasks carry the signal

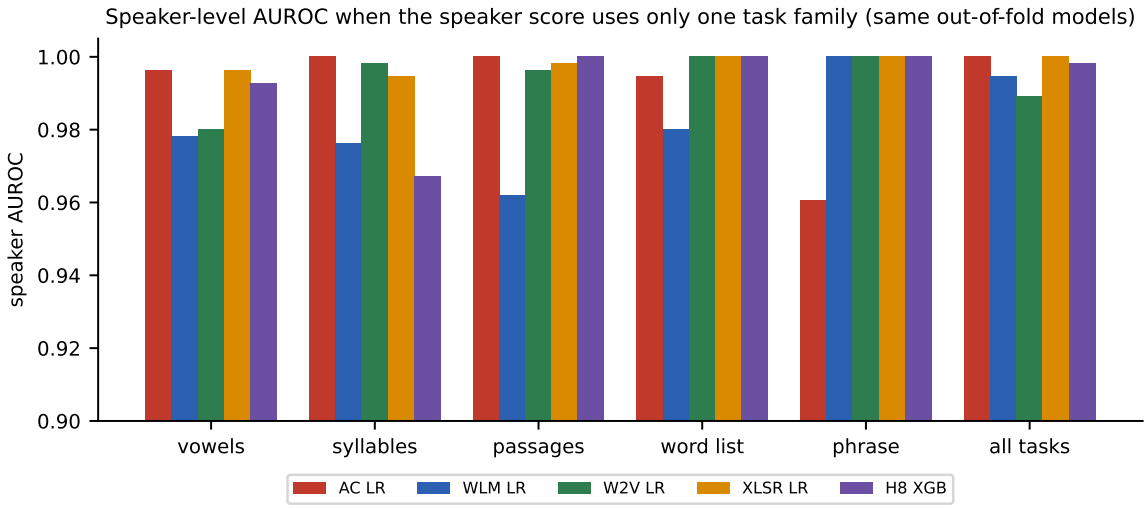


Figure 14: Speaker-level AUROC when each speaker's score is computed from one task family only (no retraining).

Table 18: Task-family analysis.

Model	Task family	Recordings	Median s	Speaker AUROC	Errors	Recording AUROC
AC LR	passages	93	65.7	1.000	1	0.976
AC LR	phrase	47	30.3	0.961	4	0.962
AC LR	syllables	100	6.3	1.000	1	0.991
AC LR	vowels	495	8.4	0.996	2	0.962
AC LR	word list	50	55.8	0.995	2	0.989
AC LR	all tasks	785	10.6	1.000	1	0.969
WLM LR	passages	93	65.7	0.962	3	0.961
WLM LR	phrase	47	30.3	1.000	2	1.000
WLM LR	syllables	100	6.3	0.976	3	0.972
WLM LR	vowels	495	8.4	0.978	3	0.950
WLM LR	word list	50	55.8	0.980	2	0.981
WLM LR	all tasks	785	10.6	0.995	1	0.959
W2V LR	passages	93	65.7	0.996	1	0.996
W2V LR	phrase	47	30.3	1.000	1	1.000
W2V LR	syllables	100	6.3	0.998	2	0.998
W2V LR	vowels	495	8.4	0.980	1	0.965
W2V LR	word list	50	55.8	1.000	0	1.000
W2V LR	all tasks	785	10.6	0.989	1	0.978
XLSR LR	passages	93	65.7	0.998	1	0.998
XLSR LR	phrase	47	30.3	1.000	0	1.000
XLSR LR	syllables	100	6.3	0.995	1	0.994
XLSR LR	vowels	495	8.4	0.996	1	0.981
XLSR LR	word list	50	55.8	1.000	0	1.000
XLSR LR	all tasks	785	10.6	1.000	1	0.985
H3 LR	passages	93	65.7	1.000	2	0.999
H3 LR	phrase	47	30.3	1.000	0	1.000
H3 LR	syllables	100	6.3	0.991	2	0.992
H3 LR	vowels	495	8.4	0.995	1	0.973
H3 LR	word list	50	55.8	1.000	0	1.000
H3 LR	all tasks	785	10.6	1.000	1	0.981
H8 XGB	passages	93	65.7	1.000	0	1.000
H8 XGB	phrase	47	30.3	1.000	0	1.000
H8 XGB	syllables	100	6.3	0.967	1	0.971
H8 XGB	vowels	495	8.4	0.993	2	0.969
H8 XGB	word list	50	55.8	1.000	2	1.000

Model	Task family	Recordings	Median s	Speaker AUROC	Errors	Recording AUROC
H8 XGB	all tasks	785	10.6	0.998	1	0.980

Every task family alone separates the groups well (speaker AUROC 0.961 or higher in every cell of the table); the 30-s phrase and the 56-s word list reach 1.000 with wav2vec2 and XLS-R, and sustained vowels have the lowest (or tied lowest) recording-level AUROC in every model. For a product this suggests a protocol of about two minutes (word list + phrase + a few vowels) - but the same caveat applies: a short recording also carries the session background.

13. The confounding problem - evidence and implications

Pre-specified diagnostic (amendment A1): train the same pipeline to recognise the **recording session inside one class**, for unseen speakers. If sessions are recognisable within a class, a PD-versus-control classifier can succeed simply by recognising sessions, because sessions and class coincide in this corpus.

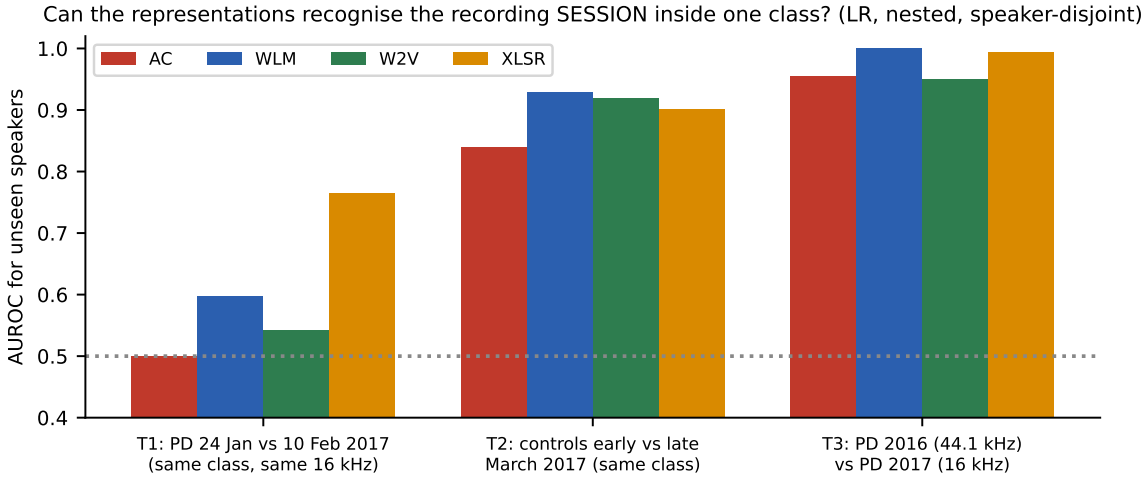


Figure 15: Session predictability within one class (unit = speaker x session; nested, speaker-disjoint).

- Controls, early versus late March 2017 (same class, same sampling rate): AUROC 0.839 to 0.929 across representations - sessions are clearly recognisable.
- PD, 2016 (44.1 kHz) versus 2017 (16 kHz): AUROC 0.950 to 1.000 - recognisable (session and resampling).
- PD, 24 January versus 10 February 2017 (same rate): AUROC 0.500 to 0.764 - near chance for three of four representations; not every session pair differs.
- Feature importance (section 10) points at background statistics (10th percentiles), and the four flagged controls (poor recording quality, reading difficulty) produce 29 of 44 false alarms.

Interpretation for investors

The current numbers cannot separate 'detects Parkinson's speech' from 'detects which session a recording came from'. Part of the signal may be genuine (spectral-shape features stay discriminative in 16-kHz-only data, and literature documents real PD speech changes [5]), but the size of the genuine effect is **unknown**. The programme in section 15 is designed to measure it.

14. Error analysis

All missed patients are one person: PD_005 is missed in 54 of 56 configurations; her recording probabilities sit just below the threshold in every task (WavLM-LR 0.19-0.59), so the models are uncertain about her as a whole. False alarms concentrate in a few controls; the four controls flagged in the corpus spreadsheet produce 29 of 44 false-alarm events. Sustained vowels have the highest recording-level error rate. In a product, recording-quality flags should abstain rather than score.

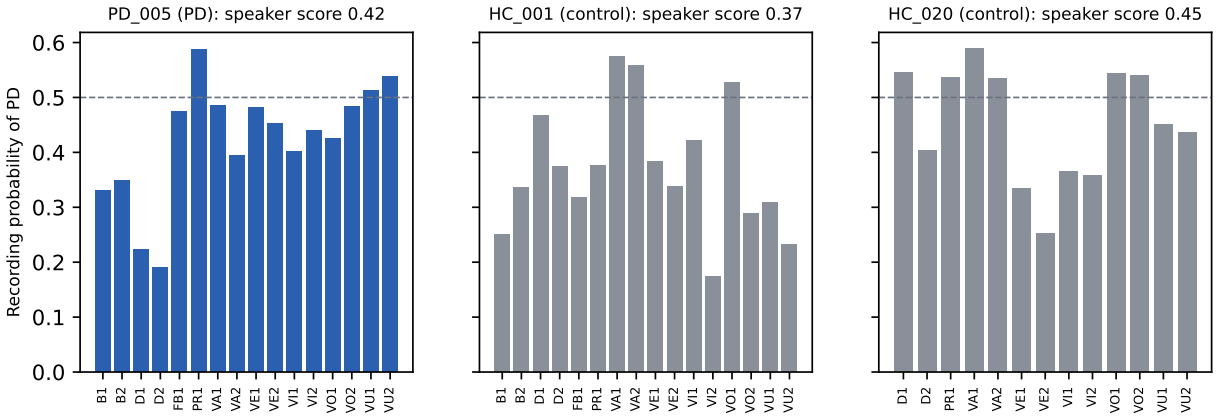


Figure 16: Recording probabilities (WavLM-LR) of the three most frequently misclassified speakers.

15. De-risking programme and validation plan

Table 19: Staged programme with decision gates (no budget figures are given; each stage is sized by the founders).

Stage	Work	Evidence produced	Gate
0 (done)	Frozen-protocol internal study and independent audit	Pipeline, 56 configurations, audit, this whitepaper	Completed
1: confound control on existing data	Background-consistency features and gate; re-analysis with background removed or matched; acquisition of a second public corpus (e.g. NeuroVoz, Castilian Spanish, 108-112 speakers [8]) for frozen external testing	A background-only model at chance on matched data; frozen model AUROC on an external corpus with pre-set success criterion (lower 95% bound $>$ 0.80)	Go/no-go 1
2: prospective recording-matched cohort	Multi-site study: PD and age/sex-matched controls recorded in the same sessions, rooms and devices, interleaved; clinical scales (MDS-UPDRS), medication state, disease duration	Sample size from the tables in section 15; pre-registered endpoints; independent statistician	Go/no-go 2
3: clinical validation	Intended-use population (for example referrals with suspected parkinsonism), differential diagnoses (essential tremor, atypical parkinsonism), multiple languages/devices	Regulatory-grade performance study; quality-management system	Regulatory submission
4: product	App, clinician report, monitoring, integration, post-market surveillance	Real-world performance monitoring	Commercial launch

Table 20: Approximate number of people needed in one class so that the exact 95% lower bound of sensitivity (or specificity) reaches a target, if the observed proportion equals the assumed true value.

Assumed true value	Required lower bound	People in that class
0.85	0.75	70
0.85	0.80	242
0.90	0.75	33
0.90	0.80	55
0.90	0.85	193
0.95	0.75	20
0.95	0.80	26
0.95	0.85	46
0.95	0.90	127

Table 21: Approximate 95% half-width of an AUROC estimate (Hanley-McNeil) for equal group sizes.

True AUROC	People per class	Half-width
0.80	50	+/- 0.087
0.80	100	+/- 0.061
0.80	200	+/- 0.043
0.80	300	+/- 0.035

True AUROC	People per class	Half-width
0.80	500	+/- 0.027
0.85	50	+/- 0.077
0.85	100	+/- 0.054
0.85	200	+/- 0.038
0.85	300	+/- 0.031
0.85	500	+/- 0.024
0.90	50	+/- 0.063
0.90	100	+/- 0.044
0.90	200	+/- 0.031
0.90	300	+/- 0.026
0.90	500	+/- 0.020

16. Regulatory pathway (to be confirmed with regulatory counsel)

- The intended product is software as a medical device (SaMD) for decision support; in the US the likely routes are 510(k) if a predicate exists or De Novo otherwise; in the EU, MDR rule 11 typically places diagnostic decision-support software in class IIa or higher.
- Precedents in speech: Aural Analytics' Speech Vitals (speech analysis, ALS) received FDA breakthrough designation, and reports describe registration of its product as a class II software device; Kintsugi has described a De Novo submission for a voice model for depression and anxiety [9]. Status changes quickly and must be checked in FDA databases.
- Required before any submission: quality-management system (ISO 13485), software life cycle (IEC 62304), risk management (ISO 14971), clinical evaluation, cybersecurity, data protection (voice is personal and health data; GDPR special-category processing).

17. Intellectual property and freedom to operate

- Own assets today: the evaluation pipeline, protocol, audit chain and the feature-level findings (copyright/know-how); no patent has been filed.
- Patent landscape: patents on speech-based diagnosis of neurological disorders exist (for example EP 4 346 558 A1, 'software-based, speech-operated and objective diagnostic tool for use in diagnosing a chronic neurological disorder' [12]); a freedom-to-operate analysis is required before product development.
- Model licences: wav2vec2-large-lv60 and XLS-R 300M are Apache-2.0; the WavLM-large model card declares no licence (to be clarified with the publisher or replaced); the Italian corpus is listed as Creative Commons Attribution [7].
- Defensibility is expected to come from proprietary, protocol-controlled clinical data and validation evidence rather than from model architecture.

18. Competitive landscape

Table 22: Selected companies working on voice biomarkers (public sources; to be verified).

Company	Focus	Source
Canary Speech (US)	Voice platform for neurological and mental-health conditions; Parkinson's listed among targets; about $22M raised (CB Insights)	[10]
Winterlight Labs (CA)	Speech-based cognitive assessment (dementia, Alzheimer's)	[10]
Sonde Health (US)	Voice-based mental, cognitive and respiratory health monitoring	[10]
Aural Analytics (US)	Speech Vitals (ALS, cognition); FDA breakthrough designation	[9]
Kintsugi (US)	Voice model for depression/anxiety; De Novo submission described	[9]
Novoic, Vocalis Health	Speech/voice AI (claims about Parkinson's for Vocalis rest on a single aggregator)	[9][10]

Defensible differentiation must come from evidence, not from the model: confound-controlled, protocol-matched clinical data; a published, auditable validation chain (frozen protocols, hash-locked artefacts, independent audits such as the one in the appendix); multi-task recording with quality gating; and low-cost CPU inference.

19. Business model hypotheses (to be tested)

- Clinical-trial services (pharma, CROs): speech as a remote, frequent digital endpoint - shortest path to revenue, lower regulatory burden if used as exploratory endpoint.
- Referral triage / tele-neurology decision support: per-test or per-site subscription; requires regulatory clearance.
- Primary-care or population screening: largest volume but lowest prevalence, hence the hardest predictive-value problem (section 2).
- Licensing of a validated scoring engine to device or platform partners.
- Unit cost of compute is negligible (section 4); value and cost sit in clinical evidence, regulatory status, integration and reimbursement.

20. Risk register

Table 23: Main risks and mitigations.

Risk	Severity	Consequence	Mitigation
Recording-condition confound	Critical	Scores may reflect sessions, not disease	Recording-matched data; background gate; background-only model must be at chance; external frozen test
Generalisation (language, device, room)	High	Performance drop outside the corpus	Multi-site, multi-device, multilingual cohorts; domain-shift monitoring
Small and narrow data	High	Wide uncertainty; no severity data	Prospective cohorts sized as in section 15; clinical scales
Label quality	Medium	Clinical PD diagnosis is 74-80% accurate [4]	Specialist-confirmed diagnoses, follow-up, differential diagnoses
Tuning instability	Medium	Saturated objectives choose arbitrary configurations	Fixed regularised defaults; external validation instead of tuning
Low prevalence in screening	High	Low positive predictive value	Enriched intended-use population; high-specificity operating point
Regulatory / reimbursement	High	Time and cost to clearance; payment unclear	Early regulatory consultation; trial-services revenue first
Privacy and consent	Medium	Voice is biometric and health data	Consent design, on-device features, data minimisation
Model and data licences	Medium	WavLM-large licence not declared on the model card; dataset CC BY	Licence clearance or replacement by Apache-2.0 encoders (wav2vec2, XLS-R perform comparably here)

21. Team and use of funds

Project team (CSE 431, topic 4a): Md. Osman Goni, Md. Mohibur Rahman, Arannamoy Mondal. The FINAL_V1 execution, audit and this document were produced with an AI assistant under the team's direction. Roles that a funded programme must add: a clinical lead (movement-disorder neurologist), a speech-language pathologist, regulatory and quality lead, data engineering, and clinical-study operations. Use of funds should follow the stages in section 15, each tranche tied to its gate; the largest cost items are expected to be prospective data collection, clinical study management and regulatory work, not computing.

22. Conclusion

The technology pipeline is sound, reproducible and cheap to run, and on the available data even simple acoustic statistics separate patients from controls almost perfectly. That is exactly why the result cannot yet be trusted: in this corpus the recording session coincides with the diagnosis, the decisive features describe the background, and sessions are recognisable within a class. The investable proposition is a disciplined de-risking programme whose first milestone - a frozen model that still separates patients from controls on recording-matched or external data while a background-only model fails - is cheap to test and decisive either way.

A. Audit summary

An independent audit (Opus 5.5, max effort) re-derived the data from raw files, re-extracted embeddings, re-implemented the acoustic features and the nested cross-validation, recomputed all metrics and bootstrap intervals, and re-executed all 70 logistic-regression runs. Verdict: no critical bug and no change to any reported

prediction or metric; two major documentation errors (the file-name number is the birth year, not the age; spreadsheet quality flags were not reported) and eight minor glitches were fixed and recorded as errata. Details: `audit_opus/OPUS_AUDIT.md`.

Table 24: Re-execution of all logistic-regression runs with single-threaded BLAS.

Condition	Same config	Max |dp|	AUROC official	AUROC re-run
AC-ALL-LR	5/5	0.0e+00	1.000	1.000
AC-BACKWARD-LR	5/5	0.0e+00	0.987	0.987
AC-FORWARD-LR	5/5	0.0e+00	0.996	0.996
H1-ALL-LR	5/5	0.0e+00	0.998	0.998
H2-ALL-LR	5/5	0.0e+00	0.996	0.996
H3-ALL-LR	5/5	0.0e+00	1.000	1.000
H4-ALL-LR	5/5	0.0e+00	0.995	0.995
H5-ALL-LR	5/5	0.0e+00	0.996	0.996
H6-ALL-LR	5/5	0.0e+00	1.000	1.000
H7-ALL-LR	5/5	4.4e-16	0.996	0.996
H8-ALL-LR	5/5	3.3e-16	0.996	0.996
W2V-ALL-LR	5/5	0.0e+00	0.989	0.989
WLM-ALL-LR	5/5	0.0e+00	0.995	0.995
XLSR-ALL-LR	5/5	0.0e+00	1.000	1.000

B. Glossary

Table 25: Glossary.

Term	Meaning
AUROC	Probability that a random patient scores higher than a random control (0.5 chance, 1 perfect)
AUPRC	Area under the precision-recall curve; chance level equals the share of patients (0.53 here)
Speaker-disjoint	No person appears in both training and test data of a fold
Nested CV	Tuning happens inside the training data of each outer fold
Confound	A variable related to both the input and the label (here: recording session)
p10 statistic	10th percentile of a frame feature over a recording - the quietest frames
Clopper-Pearson	Exact binomial confidence interval for a proportion
SaMD	Software as a medical device

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